CPSC 481
Artificial Intelligence

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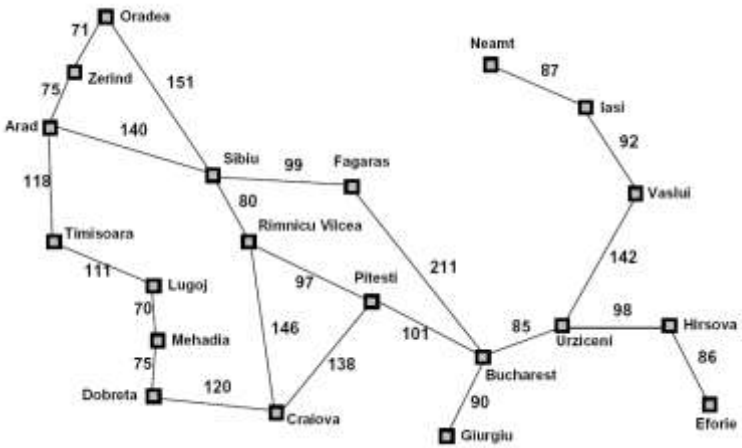
What we will cover today

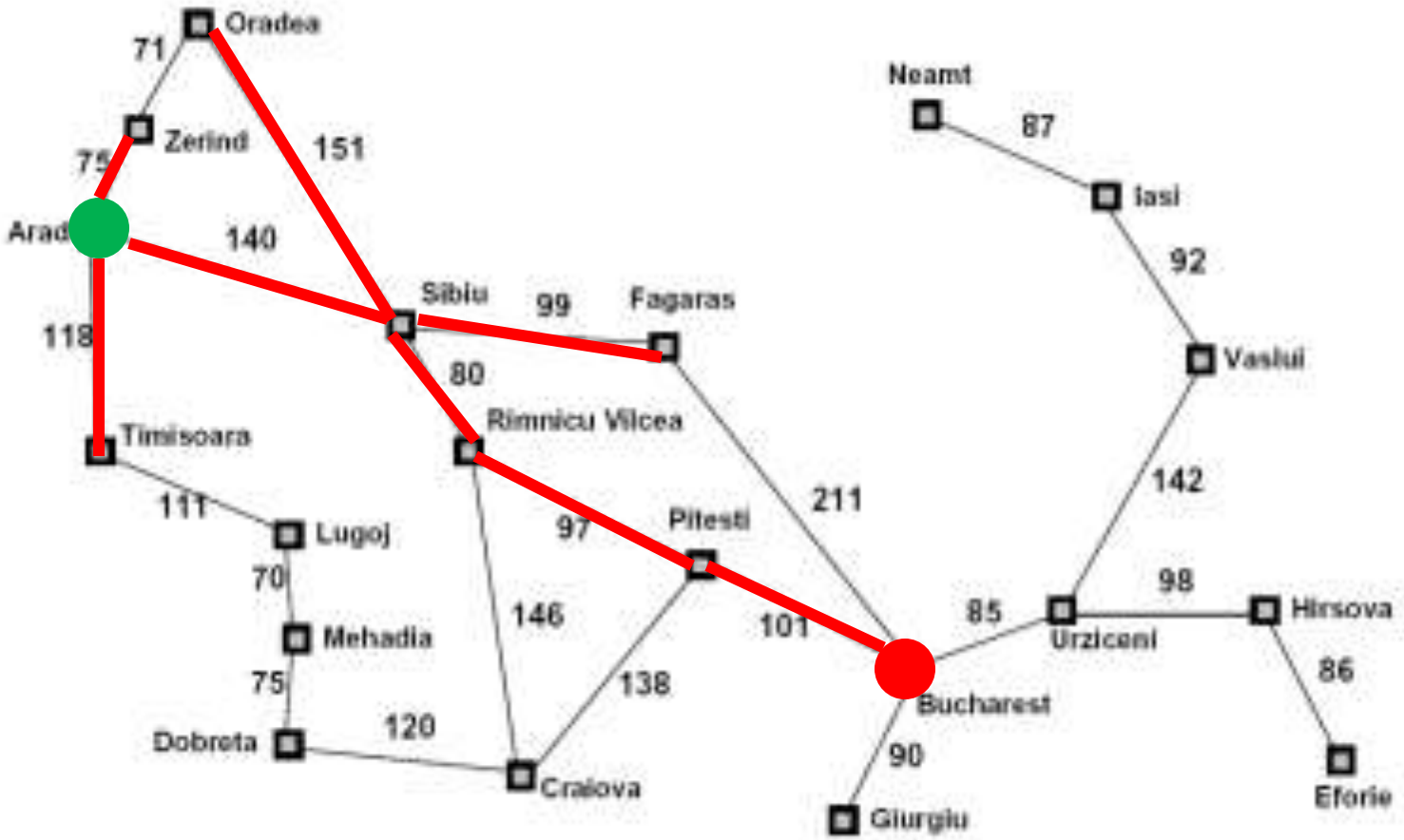
- Optimization problems and Local search algorithms
  - Hill climbing search
  - Simulated annealing
  - Genetic algorithms

Path vs. Goal

- BFS/DFS/UCS/A* all return a path to a goal state
  - The goal state is known (consider sliding tile puzzle)
  - Difficult part is getting there from the initial state
- Good:
  - Optimal
    - Systematic search of the entire state space
- Bad:
  - Slow
  - high/infeasible memory requirement
- Applications
  - Solving puzzles
  - Robot path planning
  - Parsing natural language (text -> grammar)
  - Wireless network routing



Path vs. Goal


Oradea
71
Zerind
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Arad
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Sibiu
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Fagaras
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Timisoara
Rimnicu Vilcea
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Lugoj
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Pitesti
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Mehadia
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Dobreta
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Craiova
Bucharest
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Giurgiu
Neamt
87
Iasi
92
Vaslui
142
85
Urziceni
98
Hirsova
86
Eforie

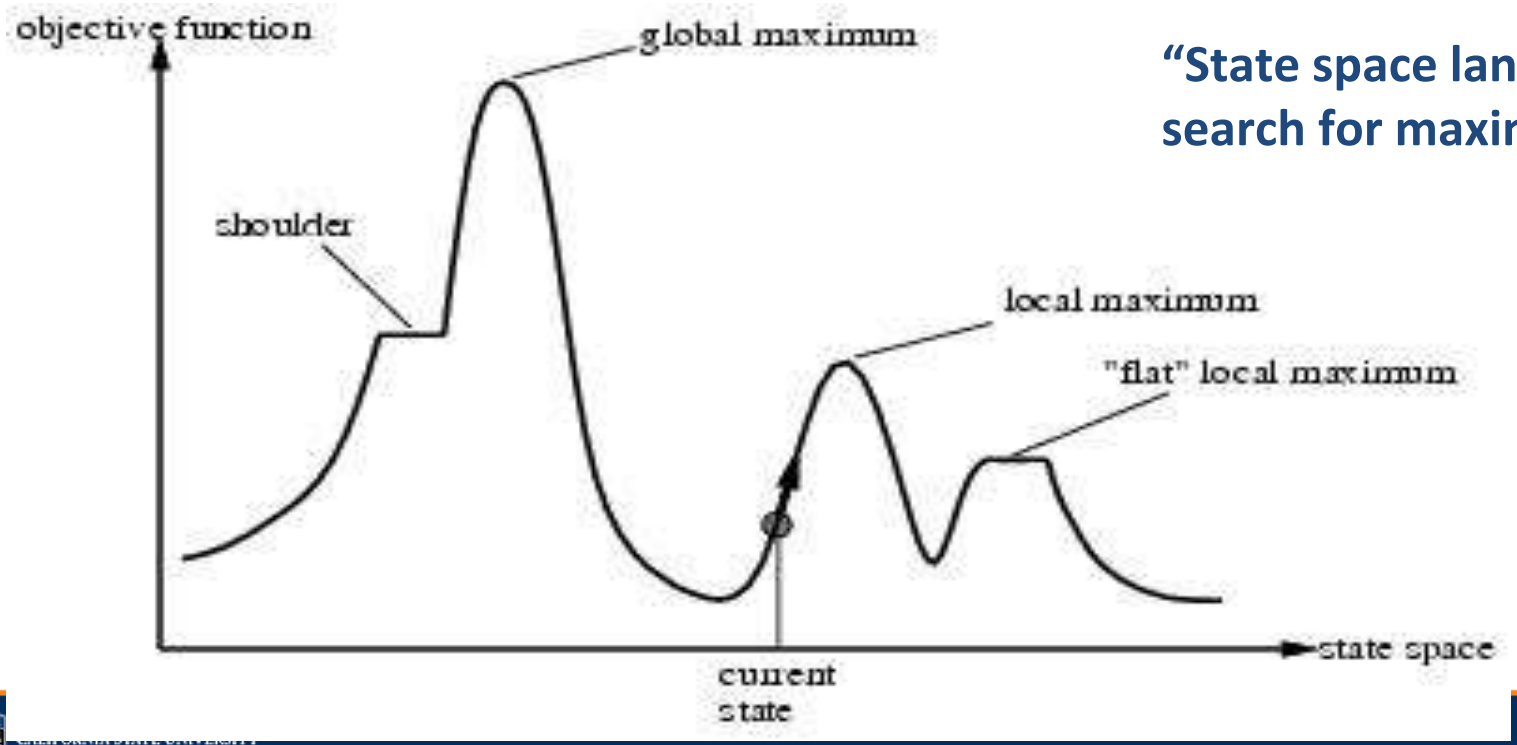
Path vs. Goal

- Optimization problems
  - Find any goal state
  - Path used to find the goal state is not important
- Applications
  - Course scheduling: fit as many classes in classrooms
  - Job scheduling
  - Constraint satisfaction problems: Sudoku
- Good:
  - Don’t worry about keeping track of paths
  - Fast
- Bad:
  - Lose optimality

5	3			7				
6			1	9	5			
	9	8					6	
8				6				3
4			8		3			1
7				2				6
	6					2	8	
			4	1	9			5
				8			7	9

Optimization problems

- Find the "**best**" state according to an objective function
  - Objective function: state → number
- Find largest value of objective function: *maximization*
- Find smallest value of objective function: *minimization*



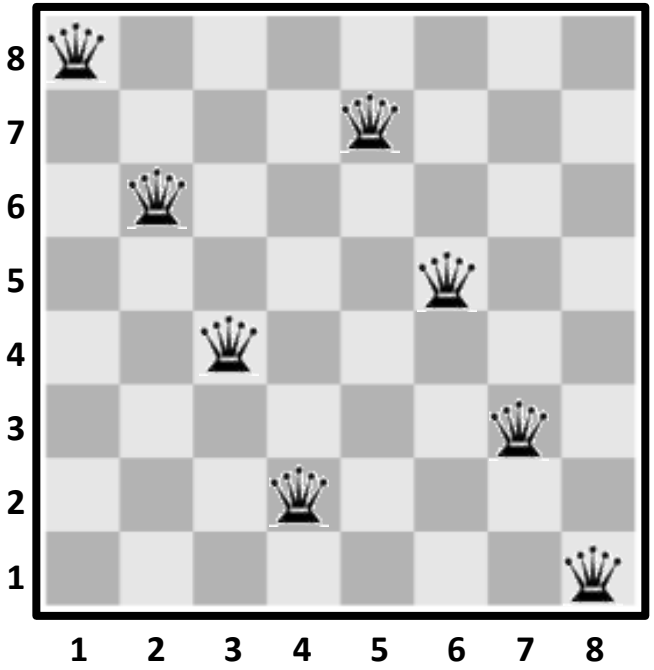
"State space landscape" for search for maximum

Local search

- Challenge:
  - Very large number of states
- Key idea:
  - Keep track of a "current state"
    - And its objective function value
  - Consider only successors of current state
    - Successor of a state: a state that is only "slightly different" from that state (also called neighbor)
    - If a successor is better than current state (higher objective function value), replace current state with this successor
  - Ignore paths
  - Repeat

Example optimization problem

- 8-queens problem
  - Place 8 queens on a chessboard such that no queen attacks another
- Size of state space?
- How to represent the state?
  - ((1,8),(2,6),(3,4),(4,2), …)
  - (8, 6, 4, 2, 7, 5, 3, 1) simpler
- Successors?
  - All states where only **one queen** is moved
- Objective function
  - Minimization: Number of pairs of queens that are attacking each other
    - For shown state?
    - Global minimum: 0
    - Is this a local minimum?
  - Maximization: Number of pairs of queens that *do not attack* each other
    - Is this a global maximum?

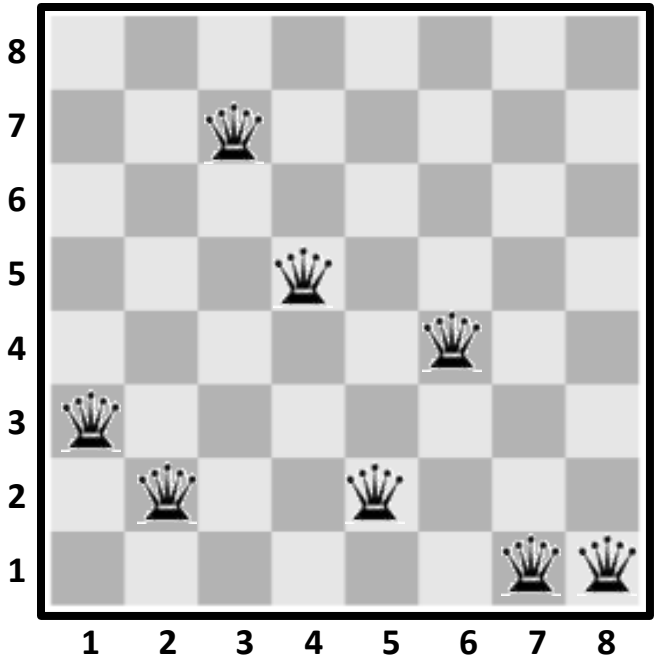


Global and local maximum

- Global maximum:
  - A state that has the highest possible objective function value in the entire state space
- Local maximum
  - A state that has the highest possible objective function value *among all its successors*
- Similarly for global and local minimum

Classwork

- Represent this board as an 8-tuple
- Calculate the objective function
  - number of attacking pairs
- Generate a few successor states
  - List the 8-tuples
- Calculate the objective function for a few successor states



Local search

HILL CLIMBING

Hill-climbing search

```
function HILL-CLIMBING( problem) return a state that is a local maximum
    input: problem, a problem
    local variables: current, a node.
                     neighbor, a node.

    current ← MAKE-NODE(INITIAL-STATE[problem])
    loop do
        neighbor ← a highest valued successor of current
        if VALUE [neighbor] ≤ VALUE[current] then return STATE[current]
        current ← neighbor
```

Algorithm sketch

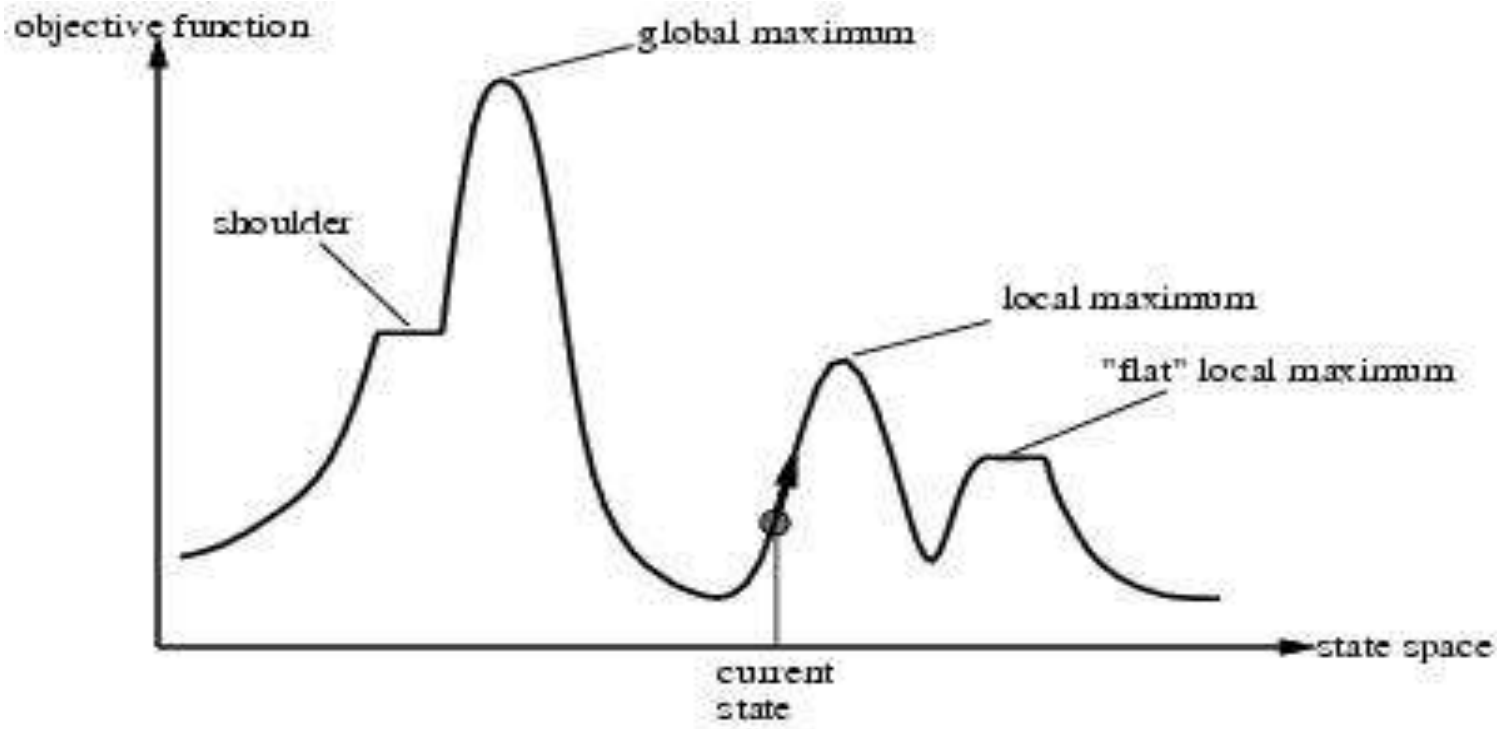
1. Expand the current state of the search.
2. Evaluate its successor states using the objective function.
3. Select the **best successor state** for further expansion.

Continue steps 1 - 3 until no better state is found

Hill-climbing search

- **Advantages:**
  - Uses very little memory
  - Can often find reasonable solutions in large or infinite (continuous) state spaces.
- But not optimal
  - Gets stuck in local maxima
  - Greedy local search

Problems of Hill-climbing



Local maxima
gets stuck when it reaches a state that has no neighboring states that are better.
Shoulder
all direct successors are not better; but a close neighbor is
Plateau
All neighboring states are not better; how far away should I keep looking for a better state?

Variations of Hill-climbing

- **Steepest ascent**
  - Version seen previously
- **Stochastic hill-climbing**
  - Of all better successors, choose one randomly
  - Intentionally not always choosing the best
  - Slower but finds better solutions in some cases
- **First-choice hill-climbing**
  - Randomly choose first better successor
  - Suitable when there are thousands of successor states
- **Random-restart hill climbing**
  - If you get stuck, start over from a random state (but note the best state seen so far)
- **Beam search**
  - Maintain a pool of *k* current states – the "beam"
  - Generate all successors of the *k* current states; keep only the best *k successors* for the new beam

Local search

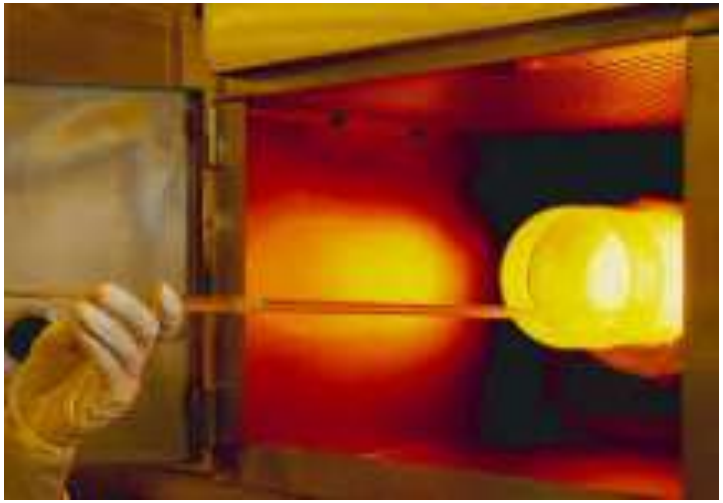
SIMULATED ANNEALING

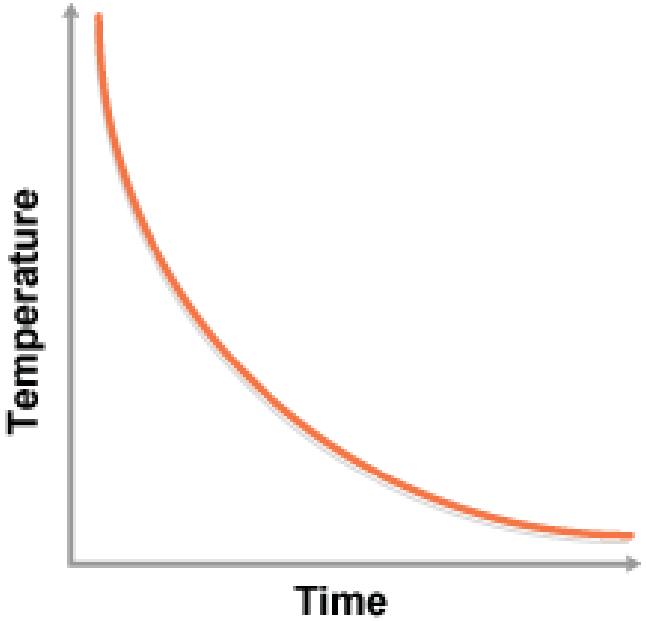
Simulated Annealing (SA)

- A hill-climbing algorithm that *never* goes downhill will get stuck in local maxima
- Better to do a random walk!
  - Will *eventually* find the global maximum
- Can we combine the two?
  - Make some downhill moves
  - But not completely randomly

Annealing

- In **metallurgy**, a technique of heating and slowly cooling metals/glass to reduce their defects and make them easier to machine





Physical Interpretation of Simulated Annealing

- simulated annealing:
  - free variables are like molecules
  - seek “low energy” (high quality) configuration
    - Note: minimization but equally applies to maximization
  - get this by slowly reducing temperature T, which controls how randomly particles move

Simulated Annealing

```
function SIMULATED-ANNEALING( problem, schedule) return a solution state
    input: problem, a problem
           schedule, a mapping from time to temperature
    local variables: current, a node.
                     next, a node.
                     T, a "temperature" controlling the probability of downward steps

    current ← MAKE-NODE(INITIAL-STATE[problem])
    for t ← 1 to ∞ do
        T ← schedule[t] # T -= 0.01
        if T = 0 then return current
        next ← a randomly selected successor of current
        ΔE ←  VALUE[next] - VALUE[current]
        if ΔE > 0 then current ← next
        else current ← next only with probability e^(ΔE/T)
```

Algorithm sketch: At each iteration, SA considers some neighbor, *next* of the current state *current*, and *probabilistically* decides whether to move from *next* to *current*.

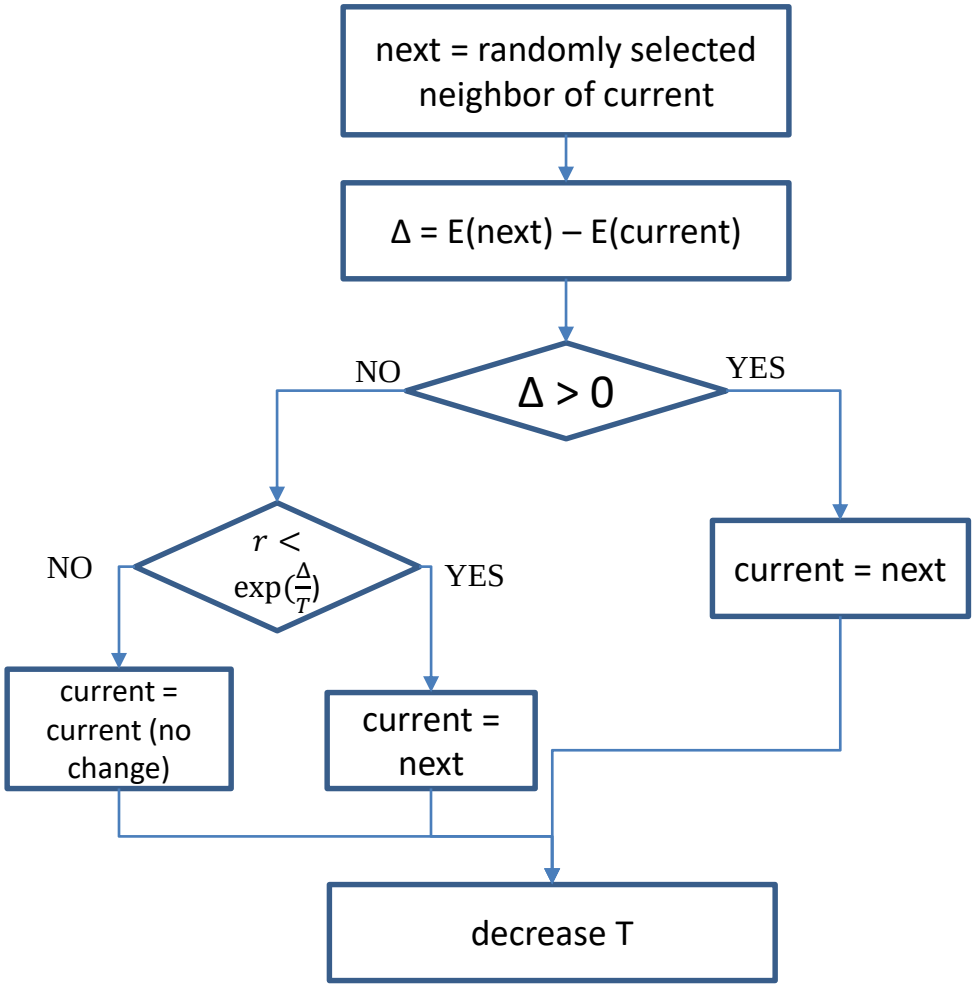
Repeat for a large number of iterations.

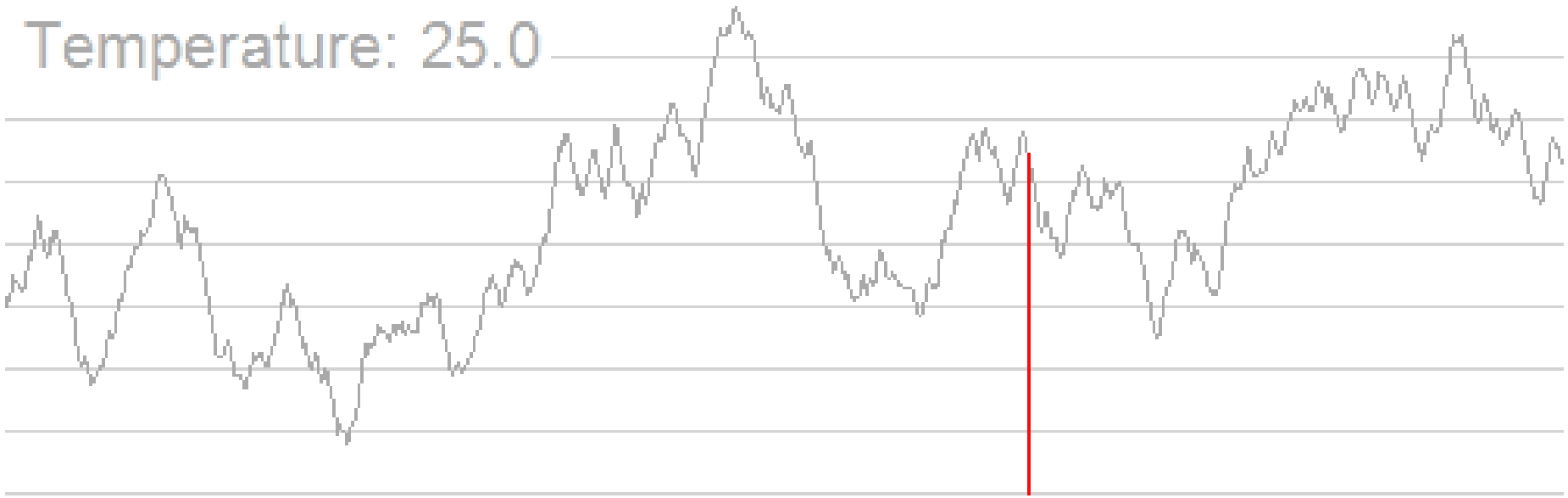
Search using Simulated Annealing

- Simulated Annealing = hill-climbing with **stochastic** moves

Basic ideas:

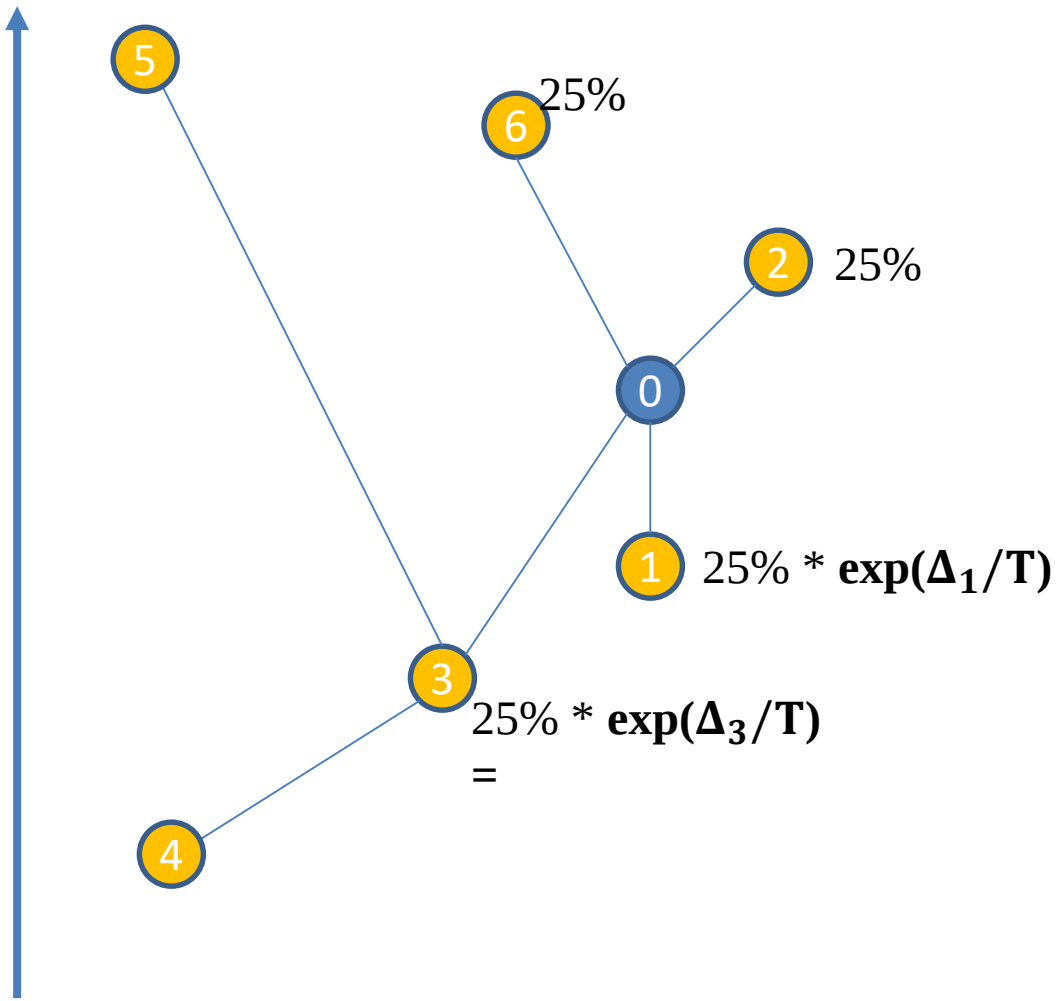
- like hill-climbing but instead of picking the best move, pick one **randomly**
- Δ = change in objective function value
- if Δ is positive, then move to that state
- Otherwise:
  - move to this state with **probability proportional to Δ**
  - thus: worse moves (large negative Δ) are executed less often
- There is always **a chance of escaping** from local maxima over time




Temperature: 25.0

Simulated Annealing example

- Current state value = 20
- Current temperature T = 1
- 3 moves/successors:
  - Objective function value = 19.1, ΔE_1 = -0.9
  - Value = 20.5, ΔE_2 = +0.5
  - Value = 15, ΔE_3 = -5
- pick a move randomly (with equal probability 1/3 = 0.333):
  - if E_2 is picked, move there (since $\Delta E_2 > 0$)
    - Prob(move to E_2) = 0.333
  - if E_1 or E_3 is picked, move there with probability = exp($\Delta E/T$)
  - move 1: prob. = exp(-0.9) = 0.406,
    - Prob(move to E_1) = 0.333 * 0.406 = 0.135
  - move 3: prob. = exp(-5) = 0.006
    - Prob(move to E_1) = 0.333 * 0.006 = 0.002


5
6
25%
2
25%
0
1
25% * $\mathbf{exp(\Delta_1/T)}$
3
25% * $\mathbf{exp(\Delta_3/T)}$
=
4

Class work

- Let current state value be 25
- Has 4 neighbors with values: 23, 20, 28, 30
- Calculate the probability of
  - Moving to 23
  - Moving to 20
  - Moving to 28
  - Moving to 30
  - Staying at 25
- Repeat for T=1 and T=10

Performance of Simulated Annealing

- T = "temperature" parameter
  - high T => probability of "locally bad" move is higher
  - low T => probability of "locally bad" move is lower
- T is decreased as the algorithm runs longer
  - *Annealing schedule*
- If the schedule is slow enough, SA finds the global optimum (with probability approaching 1)
- SA was proposed in 1983 by IBM researchers for solving VLSI layout problems (Kirkpatrick et al., Science, 220:671-680, 1983)
  - Also used for other problems: scheduling …
- Useful for some problems, but can be very **slow**
    - slowness comes about because T must be decreased very gradually to retain optimality
  - In practice **how do we decide the rate at which to decrease T?**
    - a practical problem with this method

Local search

GENETIC ALGORITHMS

Evolution by Natural Selection

- **Genome and heredity**
  - **Genome** is a complete set of chromosomes for a specie. Half of the genome (DNA) is passed from parent to child. Thus, heredity is passed through the genome.
- **Genetic variation in a population is**
  - the results of principal forces: **mutation** (various reasons) and **crossover** (sexual recombination) of genetic code (DNA)
- **Evolution is**
  - the **results** of **natural selection** (or **survival of the fittest**) over **multiple generations** through **Inheritance** (from ***reproduction***) and **genetic variation** (from ***crossover*** and ***mutation***).

What do these DNA and Evolution have to do with our problem solving approaches?

How to use Biological Knowledge in Problem Solving?

- **What we have learned from Biologists:**
  - How the information about functions and traits for an individual is **represented** and where it is **stored**.
  - **Evolutionary mechanism**
    - How is a member in a population evaluated: **Natural selection**
    - How does a population evolve? **Genetic variation**
    - What causes genetic variations?
      1. Inheritance
      2. Sexual reproduction: crossover of genes from both parents
      3. Mutation: random change
  - **Importance of diversity** in a population in evolution
    - History teaches us that all closed societies failed to survive.
- How can we use this knowledge to solve problems?
  - **Simulation** of the evolutionary mechanism
    - Genetic algorithms
    - Introduced by Holland and colleagues in 1975

GA: terminology

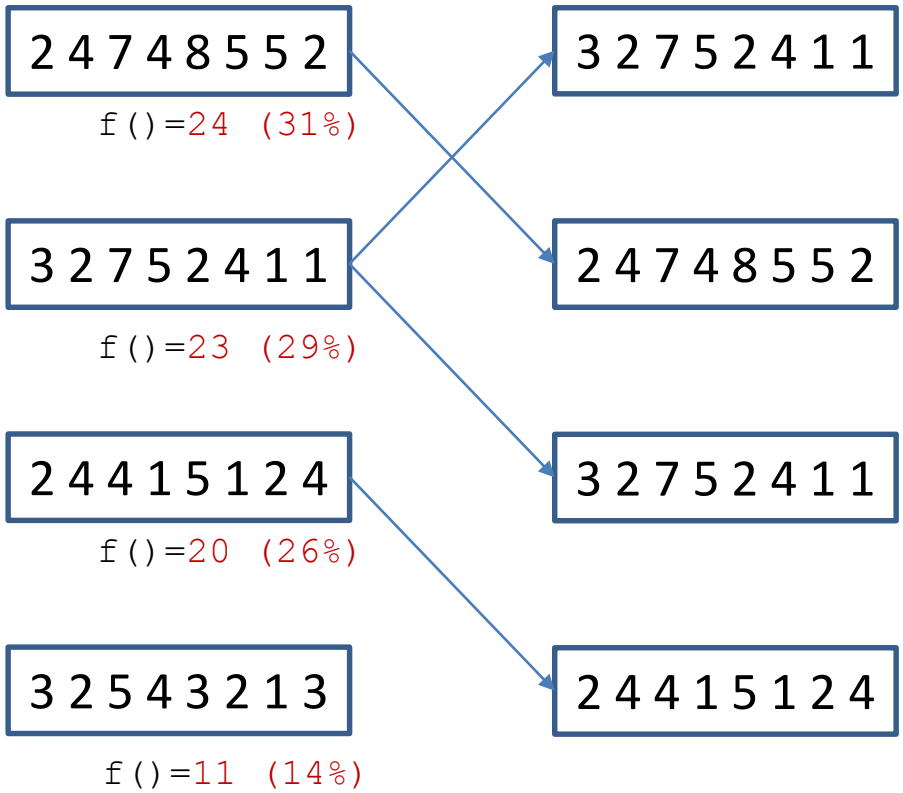
- **Population**
  - A set of candidate solutions/states
- **Crossover**
  - Combining parts of two states to create a child state
- **Mutation**
  - A random change in one bit of a state
- **Fitness function**
  - A numeric measure of the relative goodness of a state

Genetic Algorithm

```
function GENETIC-ALGORITHM(population, fitness) returns an individual
  repeat
      weights ←WEIGHTED-BY(population, fitness)
      population2 ← []
      for i = 1 to SIZE(population) do
          parent1 , parent2 ←WEIGHTED-RANDOM-CHOICES(population,weights, 2)
          child ←REPRODUCE(parent1, parent2 )  // crossover
          if (small random probability) then child ←MUTATE(child)
          add child to population2
      population ←population2
  until some individual is fit enough, or enough time has elapsed
  return the best individual in population, according to fitness
```

```
function REPRODUCE(parent1, parent2 ) returns an individual
      n←LENGTH(parent1)
      c←random number from 1 to n
      return APPEND(SUBSTRING(parent1, 1, c), SUBSTRING(parent2, c + 1, n))
```

8-queens problem

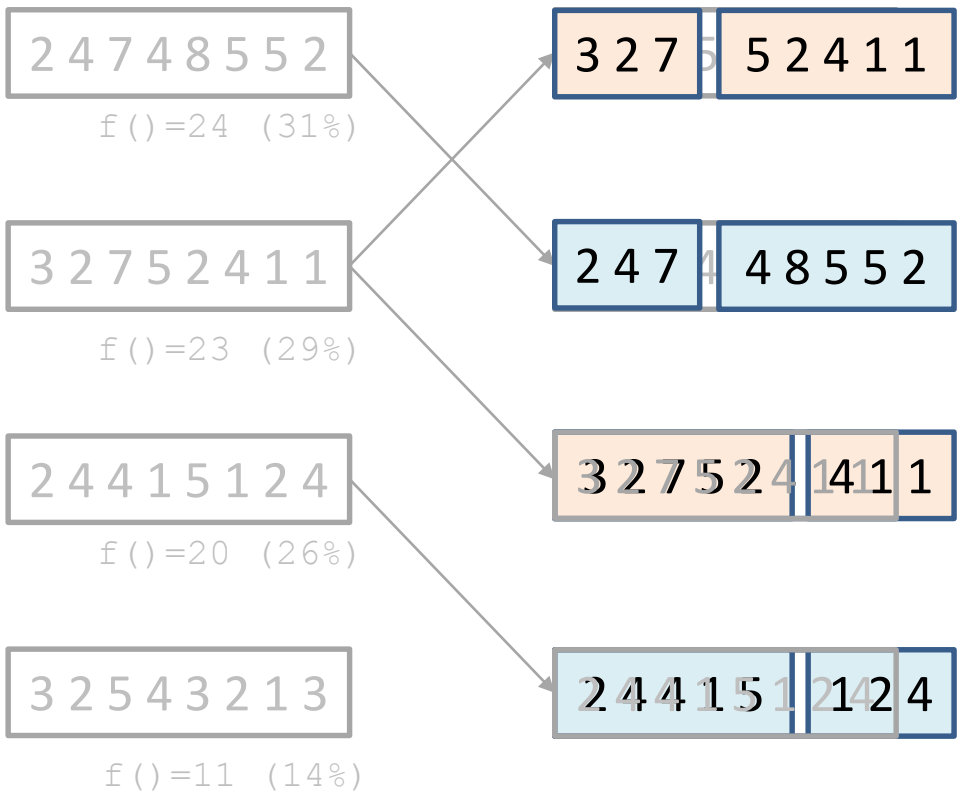


(a) Initial population

(b) Fitness function

(c) Selection

8-queens problem



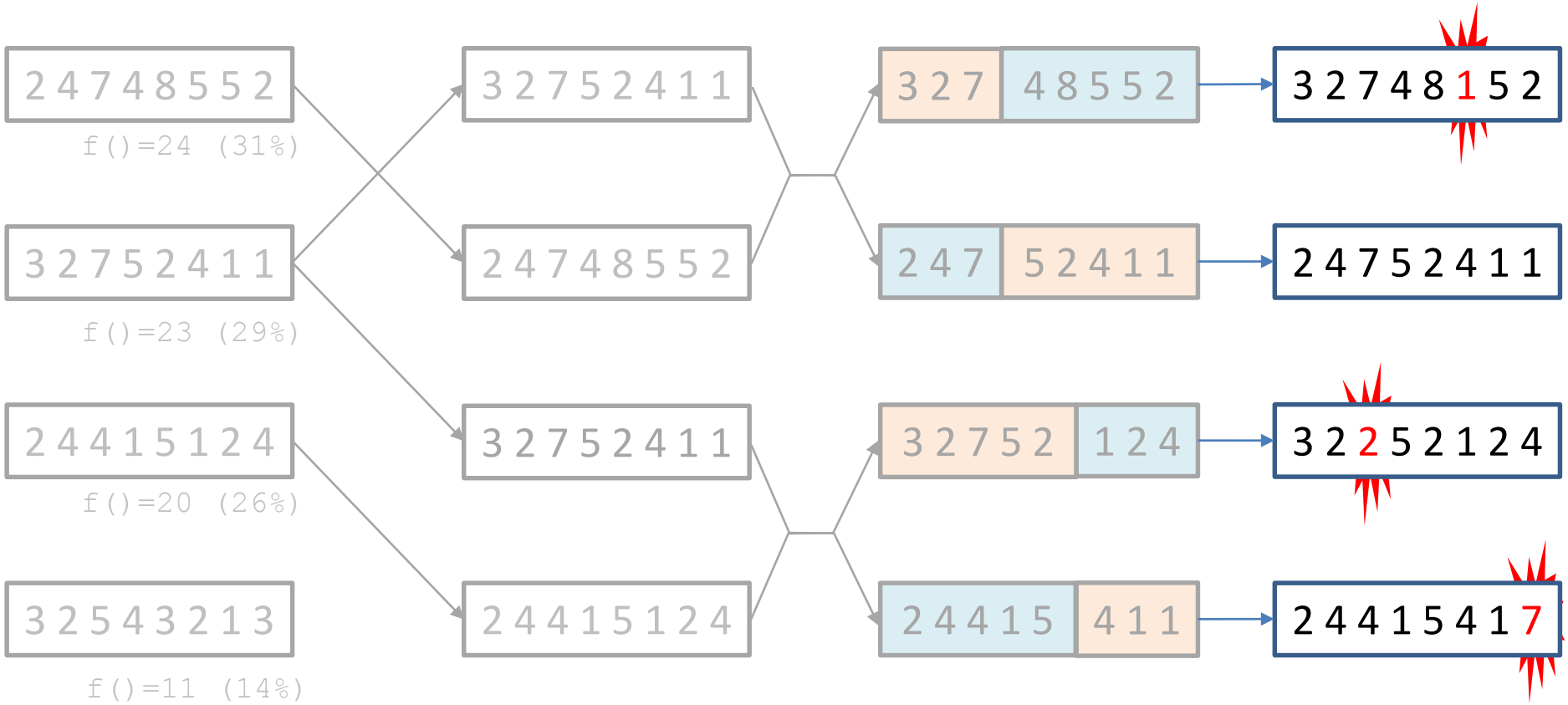
(a) Initial population

(b) Fitness function

(c) Selection

(d) Crossover

8-queens problem



(a) Initial population

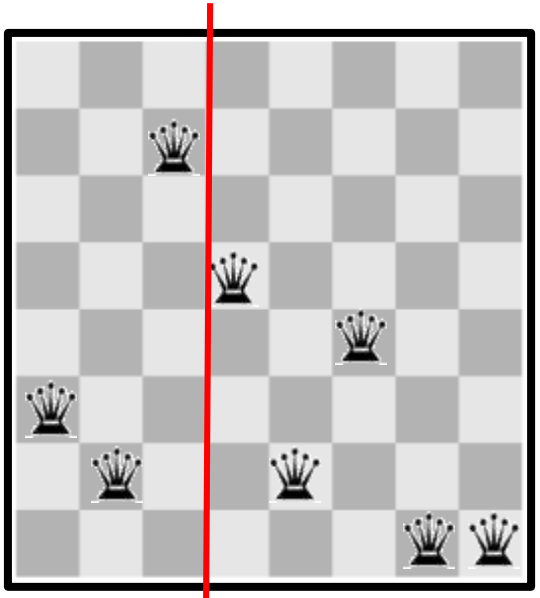
(b) Fitness function

(c) Selection

(d) Crossover

(e) Mutation

8-queens problem: crossover



[3 2 7 | 5 2 4 1 1]

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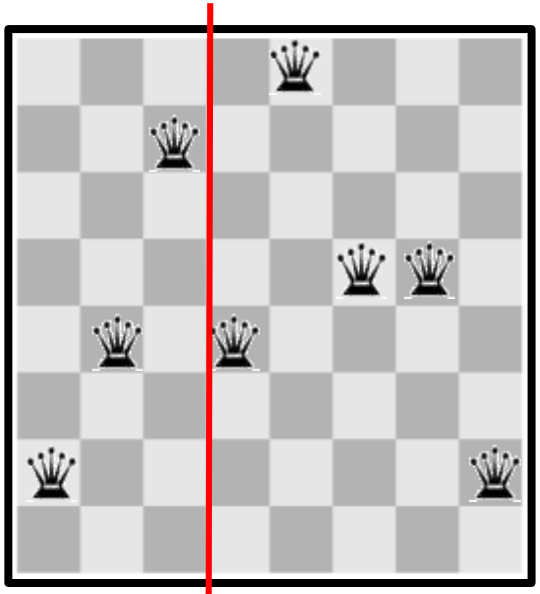

[2 4 7 | 4 8 5 5 2]

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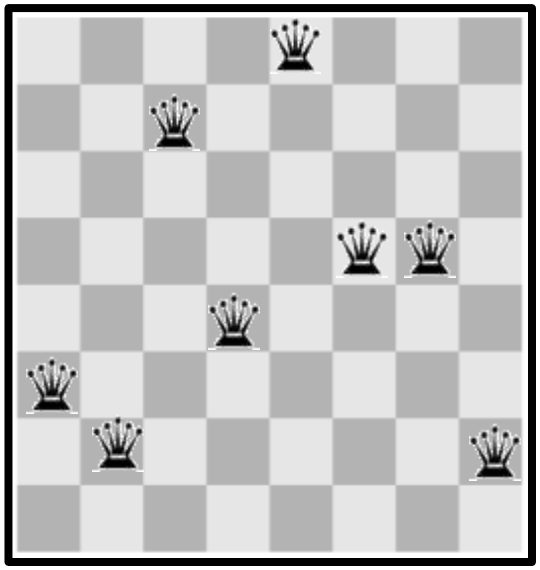

= [3 2 7 4 8 5 5 2]

Classwork: crossover

- Show a few other "children" of these two parent states
  - **[3 2 7 5 2 4 1 1]**
  - **[2 4 7 4 8 5 5 2]**

Setting up a GA for a specific problem

- Representation for a candidate solution (DNA)
  - Use a **fixed length string** (binary bit string).
- Initial population of candidate solutions (chromosomes)
  - Create randomly (each candidate solution can be a random binary string.)
- Fitness criteria (measure for survival of fitness)
  - Define a fitness function f(x) to evaluate each candidate solution.
- Creating next population with genetic variations (evolution)
  - Select survived candidate solutions.
  - Apply genetic operators to the selected solutions to produce improved solutions in next generation (offsprings with modified DNA).
- Terminating conditions and other parameters
  - population size, probability of crossover, probability of mutation, number of generations, etc.

Performance of GA

- Have been used in many optimization problems
  - Scheduling
  - Traveling Salesman Problem
- Works well for problems where *parts of a good candidate* should be left unchanged
- Disadvantages
  - Representation is not obvious and impacts how well it does
  - Several hyper-parameters:
    - Size of population, mutation probability, number of iterations, terminating condition, …

References

- George F. Luger, Artificial Intelligence: Structures and Strategies for Complex Problem Solving, 6th edition, Addison Wesley, 2009. **Chapters 4.1.1, 12.1**.
- Russel and Norvig, Artificial Intelligence: A Modern Approach, 3rd edition, Prentice Hall, 2010. **Chapter 4-4.1**